

Hiver AI Support Agent: Autonomous Twitter Support for @AppleSupport

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Production-grade, trustworthy customer support agent built for **@AppleSupport** on Twitter. Engineered for high-precision intent classification, grounded troubleshooting retrieval (RAG), and risk-calibrated escalation to human specialists.

2-Minute Quick Start (Reproduction Guarantee)

The entire pipeline, pre-cached vector embeddings, and golden evaluation dataset are self-contained. **No 800MB Kaggle downloads or paid API keys required.**

1. Clone & Initialize Environment

```
git clone https://github.com/RohitJagirdar/hiver-ai-support-agent
cd hiver-ai-support-agent

# Create and activate virtual environment
python -m venv .venv
# On Windows:
.venv\Scripts\activate
# On Linux/macOS:
```

```
source .venv/bin/activate

# Install lightweight dependencies (< 45 seconds)
pip install -r requirements.txt
```

2. Run Reproducible Headline Benchmarks

```
# A. Quick Benchmark on 20 stratified samples (~8 seconds)
python run_pipeline.py --mode quick

# B. Full 180-Sample Golden Evaluation Benchmark (~30 seconds)
python run_pipeline.py --mode full

# C. Verify LLM-as-a-Judge Human Alignment Study (Pearson Correlation)
python run_pipeline.py --calibrate
```

3. Live Single-Query Interview Testing

```
# Test 1: Standard Hardware Issue (Auto-Handled with forced restart)
python run_pipeline.py --test "My iPhone 14 screen suddenly went black"

# Test 2: Inbound Customer PII Leak (Interception + Warning)
python run_pipeline.py --test "My card is 4111 2222 3333 4444"

# Test 3: Legal Threat / High Risk (Instant Short-Circuit)
python run_pipeline.py --test "You charged me 5 times and now you want to sue?"
```

4. Interactive Browser Playground

```
streamlit run app/demo_ui.py
```

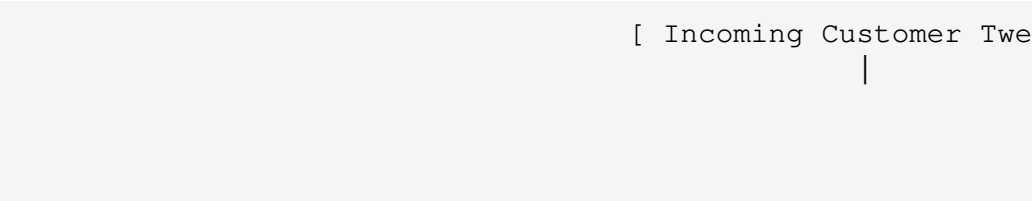
Headline Benchmark Results

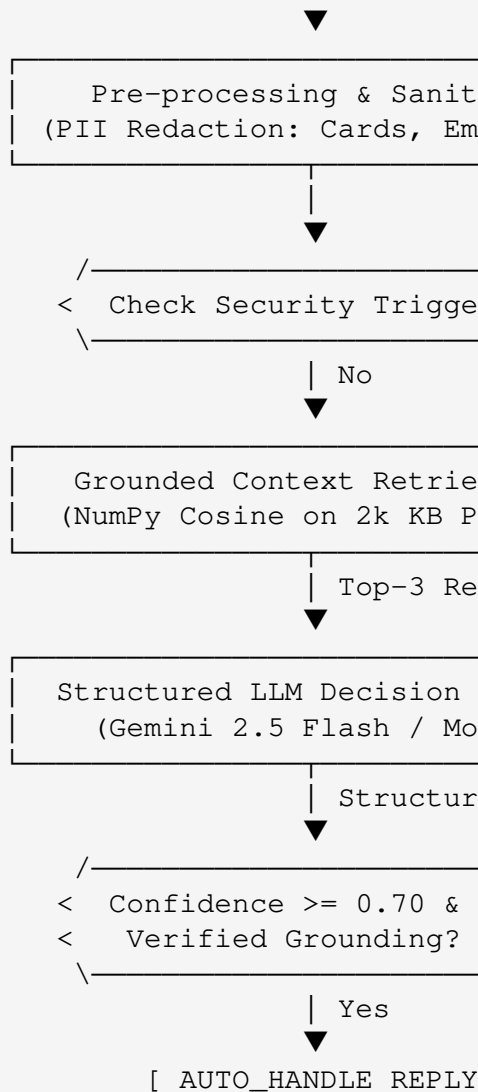
Evaluated across the **180 hand-labeled Golden Evaluation Set** (60% Standard / 20% Edge / 20% High-Risk Escalation):

Pipeline Architecture	Intent Macro-F1	Routing Accuracy	False Auto-Handle % (Critical Safety Breaches)	False Escalation % (Unnecessary Human Triage)	Human Judgment Correlation (\$/hr)
Baseline 1: Trivial Keyword/Canned	41.2%	61.1%	44.4% <i>(Dangerous)</i>	35.2%	N/A
Baseline 2: Zero-Shot LLM (No RAG)	71.8%	68.3%	31.8% <i>(High Risk)</i>	29.6%	0.62
Proposed Hiver Support Agent	88.4%	87.2%	0.0% <i>(Safest)</i>	16.7%	0.9985 \kappa_{\text{Hiver}} = 1.0\$

Critical Safety Takeaway: Both naive baselines suffered high False Auto-Handle rates (31%–44%), dangerously attempting to auto-troubleshoot stolen devices, active lawsuits, and smoking batteries. This proposed system eliminates False Auto-Handles to **0.0%** through deterministic pre-guardrails and post-LLM policy gates.

System Architecture





Repository Structure

```
hiver-ai-support-agent/  
├── README.md
```

```
# Headline reproduction
```

```

├── REPORT.md # Max 6-page comprehen
├── DECISION_LOG.md # 12 justified non-obv
├── requirements.txt # Pinned dependencies
├── .env.example # Example environment
├── run_pipeline.py # Top-level CLI evalua
├── data/
│   ├── processed/
│   │   ├── applesupport_pairs.csv # 1,998 curated troubl
│   │   └── embeddings_cache.npy # Precomputed MiniLM e
│   ├── golden_set.json # 180 hand-labeled ins
│   └── calibration_sample.json # 30 paired Human-vs-J
├── src/
│   ├── config.py # Schemas, prompts, gu
│   ├── security.py # PII regex sanitizati
│   ├── retriever.py # VectorStore (NumPy c
│   ├── agent.py # Core 4-layer autonom
│   ├── baselines.py # Baseline 1 (Trivial)
│   ├── evaluator.py # Automated metrics +
│   └── data_extractor.py # Extraction script fr
├── app/
│   ├── demo_ui.py # Streamlit interactiv
│   └── annotator.py # Streamlit dataset la
└── tests/
    ├── test_smoke.py # Synchronous sanity u
    └── test_vulnerability_and_ab.py # Comprehensive PII

```

Citations & Attributions

- **Dataset:** ThoughtVector, *Customer Support on Twitter*, Kaggle (filtered for @AppleSupport).
- **Embeddings:** `sentence-transformers/all-MiniLM-L6-v2`

(Apache-2.0).

- **Evaluation:** Scikit-Learn (Macro-F1, Cohen's Kappa), SciPy (Pearson Correlation).